

YUEZHEN DONG

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RELEVANT SKILLS

Languages	Python, Java, TypeScript/JavaScript, C/C++, SQL, Ruby, Bash
ML/AI	PyTorch, scikit-learn, OpenAI API, Gemini API, RAG, LLM Agents, Prompt Engineering
ML Methods	Supervised learning, classification, model evaluation, causal inference, A/B testing
Frameworks	FastAPI, Flask, React, Node/Express, Ruby on Rails, Spring Boot, GraphQL
Infrastructure	Docker, PostgreSQL, AWS, Git, Linux

EDUCATION

University of Waterloo – Faculty of Engineering	Expected Graduation MAY 2028
BASc in Computer Engineering	Waterloo, ON, CAN

ACHIEVEMENTS

IBDP Graduate CaYPT (Canadian Young Physicists' Tournament) National Champion IYPT (International Young Physicists' Tournament) Bronze Medalist FIRST Robotics Provincial Semi-Finalist

WORK EXPERIENCE

Meta Production Engineering Fellow JUN 2026 – SEP 2026
Linux, Python, Bash, Docker, CI/CD, Networking, Monitoring, Site Reliability Engineering *Meta*

- Selected to work with **Meta engineers** on production-scale infrastructure, developing SRE tooling for automated deployments, service health monitoring, log-based diagnostics, failure analysis, and Linux systems reliability.

Software Engineering Intern MAY 2026 – AUG 2026
Python, TypeScript, React Native, Ruby on Rails, RAG, LLM, OAuth, BLE *Shopify*

- Contributed to **agentic AI tooling infrastructure** by implementing **RAG-based context retrieval** with embedding reranking and prompt orchestration pipelines, reducing irrelevant context fed to the LLM by **~40%** and improving AI-generated code acceptance rate in developer review.
- Built **Quick Device Activation** for Shopify POS using **BLE** pairing, QR/manual-code fallback, and **OAuth Device Authorization** – reducing merchant device setup time from **~5 minutes to under 30 seconds**.
- Instrumented login-conversion metrics across **React Native** and **Rails** using feature flags, enabling A/B testing of activation flows and identifying a drop-off step that accounted for **35% of incomplete activations**.

Software Engineering Intern SEP 2025 – DEC 2025
React, TypeScript, Ruby on Rails, GraphQL *Shopify*

- Owned a **staff assignment modal** in Retail Admin end-to-end – built React components and designed **GraphQL** queries with server-side pagination, reducing staff search query response time by **~60%**.
- Shipped full-stack features across **6 systems** (Admin Web, POS Mobile, Shopify Core, Business Platform, Cloud Sync Streamer, POS Channel), maintaining backward-compatible data contracts and achieving **zero regression** on launch.

Web Developer Coop JAN 2025 – APR 2025
Java, Spring Boot, Maven, Apache POI, Git *AGF Investments*

- Built a **Java/Spring Boot** ScoreCard application with XLSX streaming via **Apache POI**, configurable transformation pipelines, and input validation, automating report generation across **4 departments** and saving **20+ staff hours/week** previously spent on manual spreadsheet processing.

PERSONAL PROJECTS

Multimodal Agent Evaluation Sandbox

Python, PyTorch, Gemini API, FastAPI, React, PostgreSQL, Docker

- Built a **multimodal benchmark for AI agents** evaluating text, image, code, and tabular reasoning through **tool-calling**, **RAG**, sandboxed Python execution, and verifier-based scoring – agents evaluated on this benchmark achieved **23% higher accuracy** on unseen task variants compared to a zero-shot baseline.

AI Merchant Inventory Assistant

Python, TypeScript, OpenAI API, scikit-learn, React, Node/Express, SQLite

- Built an **LLM-powered inventory assistant** with an **agentic tool-routing pipeline** that classifies merchant intent and dispatches to structured SQLite queries across 5 inventory workflows, achieving **89% intent classification accuracy** on a 200-query evaluation set.